

Corporate Insolvency: Predictive Modeling and Unsupervised Subtype Analysis

The Indian corporate landscape has witnessed spectacular financial collapses, from Kingfisher Airlines to the Amtek Auto group. Traditional financial screeners often miss these events because they rely on easily manipulated metrics like Net Income or top-line revenue.

This project builds a predictive insolvency model focused on forensic accounting indicators. Rather than just looking at negative profitability, we are testing specific failure signatures:

1. **The Intangible Asset Illusion:** Distressed companies frequently inflate their balance sheets using heavily impaired “intangible assets” or “goodwill” to secure leverage, masking an underlying negative net worth.
2. **The Working Capital Trap:** A severe divergence between reported EBITDA (paper profit) and Operating Cash Flow (actual liquidity), usually driven by uncollectible receivables.
3. **The Debt Avalanche:** Capital structures overly reliant on short-term borrowings to fund long-term, depreciating assets, leading to sudden liquidity crises when credit markets tighten.

The Dataset

We are utilizing `insolvency_raw.csv`, a dataset containing ~900 Indian listed companies. Crucially, the “bankrupt” class ($\text{Target} = 1$) features a curated list of corporate insolvencies gathered through standard financial platforms (such as Yahoo Finance, Screener, Investing.com and Annual Reports). This data is fortified by rigorous manual vetting of company annual reports from the fiscal years immediately preceding their collapse, capturing their financial state at the exact moment the terminal distress became mathematically irreversible.

Data Gathering Notes

The raw dataset is built using a dual-stream pipeline that independently captures active and distressed companies before merging them into a unified schema.

Active Companies ($\text{Target} = 0$)

- **Sourcing & Extraction:** Data is pulled via the Yahoo Finance API. To maintain transparency, rows with missing financial periods are not silently dropped; instead, they are preserved and explicitly flagged with a retrieval status.

Bankrupt Companies (Target = 1)

- **Source:** Ground-truth terminal tickers are deterministically aggregated from the NSE's periodic lists of delisted, suspended, and non-compliant (Z-category) companies.
- **Process:** The ingestion engine maps these tickers to historical NSE archives and company filings to locate the exact Annual Report PDF published immediately prior to the insolvency event. The logic strictly prioritizes consolidated statements over standalone statements to ensure hidden subsidiary debts and off-balance-sheet liabilities are captured.
- **Extraction:** Due to the complex, non-standardized, and frequently scanned nature of Indian corporate filings, a custom LLM-based parsing agent is deployed to extract the raw quantitative metrics. This agent operates with automated retry mechanics to handle transient API failures or complex table layouts, attaching a detailed extraction log and status marker to every processed row.

Schema Consolidation & Merging The two streams are mapped into a standardized format. Duplicate tickers—which often occur when a company triggers multiple exchange warnings over several quarters—are resolved by evaluating row completeness and temporal proximity to the actual collapse, keeping only the most accurate terminal year for the master file.

Dataset Features

Metadata Fields

- **ticker:** Stock ticker symbol (e.g., RELIANCE, TCS)
- **company_name:** Legal company name
- **sector:** Industry sector
- **industry:** Industry classification
- **fiscal_year:** Financial year of data
- **currency:** Currency of financial values (INR)
- **source_url:** Where the data was sourced from
- **extraction_status:** Data collection status
- **extraction_error:** Any errors during data extraction

Financial Features (Key Indicators)

- **market_cap:** Market capitalization
- **total_debt:** Total debt outstanding
- **intangible_assets:** Goodwill and intangible assets

- `cash_and_equivalents`: Cash on hand
- `current_liabilities`: Short-term obligations
- `operating_cash_flow`: Cash generated from operations
- `ebitda`: Earnings before interest, tax, depreciation, amortization
- `interest_expense`: Interest paid on debt
- `net_income`: Profit/loss for the period
- `total_assets`: Total resources controlled by company

Target Variable

- `target`: 1 (Insolvent/Bankrupt), 0 (Healthy/Solvent)

Environment Setup

Environment initialized.

Data Cleaning & Imputation

Raw shape = (1379, 21)

<class 'pandas.DataFrame'>

RangeIndex: 1379 entries, 0 to 1378

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	ticker	1379 non-null	str
1	company_name	1379 non-null	str
2	sector	872 non-null	str
3	industry	872 non-null	str
4	business_summary	874 non-null	str
5	target	1379 non-null	str
6	market_cap	887 non-null	str
7	total_debt	1064 non-null	str
8	intangible_assets	1014 non-null	str
9	cash_and_equivalents	1100 non-null	str
10	current_liabilities	1080 non-null	str
11	operating_cash_flow	1020 non-null	str
12	ebitda	1064 non-null	str
13	interest_expense	1080 non-null	str
14	net_income	1093 non-null	str
15	total_assets	1083 non-null	str
16	fiscal_year	1199 non-null	str

```

17  currency          1379 non-null  str
18  source_url        1332 non-null  str
19  extraction_status  1379 non-null  str
20  extraction_error   275 non-null  str
dtypes: str(21)
memory usage: 226.4 KB

```

None

```

Target distribution
target
0      0.66
1      0.34
target  0.00
Name: proportion, dtype: float64
Duplicate-ticker rows = 341

```

	ticker	company_name	extraction_status
679	ABGSHIP	ABG Shipyard	Success_Manual
1259	ABGSHIP	ABG Shipyard	pdf_llm_consolidated
780	AGRITECH	AGRI-TECH (INDIA) LIMITED	yfinance
1068	AGRITECH	Agri-Tech (India) Limited	pdf_llm_standalone
398	AKSHOPTFBR	AKSH OPTIFIBRE LIMITED	yfinance

Cleaning Execution

From this point onward, we apply transformations to fix the issues diagnosed in cells 6 to 10.

Cleaning goals: - Preserve key identifier columns (**ticker**, **company_name**, **target**) - Standardize text and URL formats - Convert numeric fields from comma-formatted strings to numeric dtype - Resolve duplicate tickers using data completeness - Handle missing values in financial columns - Validate dataset integrity before EDA

```

Cleaned shape = (854, 21)
Rows removed (metadata_only) = 180
Rows removed (>50% missing financial) = 106
Rows removed (strict financial completeness) = 100
Total rows removed = 525

```

Cleaning summary

	metric	before	after
0	rows	1379	854
1	columns	21	21
2	duplicate_ticker_rows	341	0
3	financial_missing_values	2813	0
4	unique_tickers	1181	854

Saved cleaned file = insolvency_clean.csv

Exploratory Data Analysis

1: Dataset Snapshot

Quick sanity checks after cleaning: shape, class balance, extraction status mix, and basic numeric summary.

Cleaned shape = (854, 21)

Target counts

target

0 685

1 169

Name: count, dtype: int64

Extraction status counts

extraction_status

YFINANCE 727

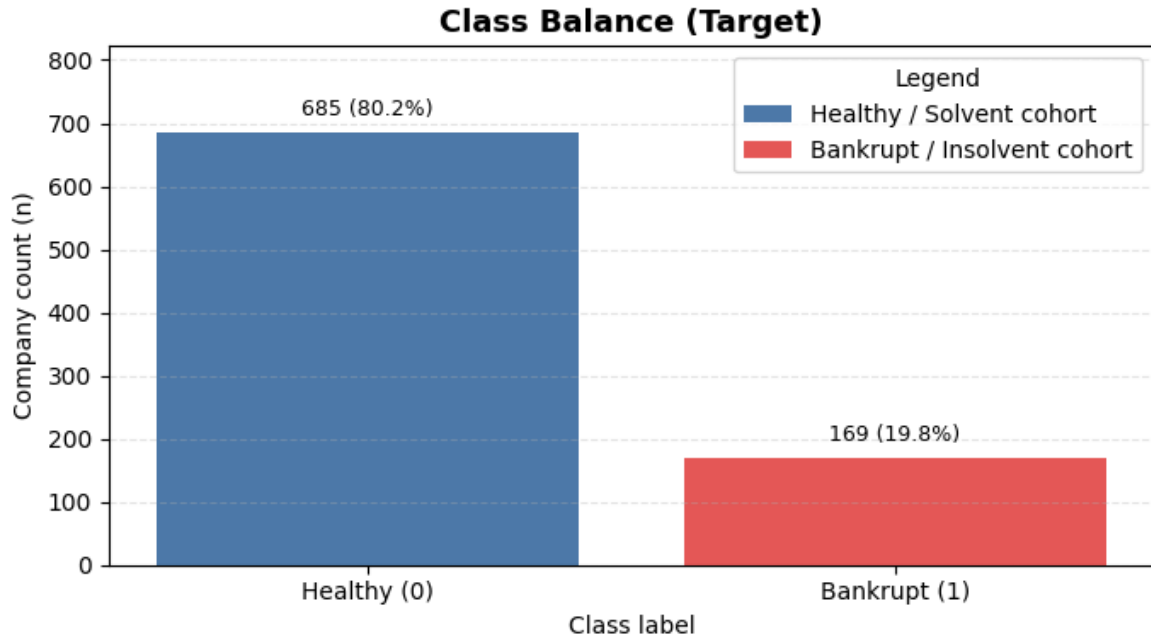
SUCCESS_MANUAL 73

PDF_LLM_CONSOLIDATED 38

PDF_LLM_STANDALONE 16

Name: count, dtype: Int64

	count	mean	std	min	25%	50%
total_debt	854.00	65606100546.91	304500410964.70	0.00	869550000.00	304500410964.70
intangible_assets	854.00	18000708772.00	161736147532.04	0.00	5510000.00	18000708772.00
cash_and_equivalents	854.00	21151761493.60	101189188139.89	0.00	313536569.00	21151761493.60
current_liabilities	854.00	60749289173.63	237041065594.47	0.00	2865739250.00	60749289173.63
operating_cash_flow	854.00	15577501904.18	91863190397.21	-621132500000.00	180703250.00	15577501904.18
ebitda	854.00	24438689805.46	99921520542.17	-68000000000.00	616244750.00	24438689805.46
interest_expense	854.00	4241175584.47	18809440133.10	-236095000.00	73940000.00	4241175584.47
net_income	854.00	10258675893.97	45237888404.91	-273834000000.00	89897750.00	10258675893.97
total_assets	854.00	205914065807.74	906402554577.49	20340000.00	10331559250.00	205914065807.74



Scale: y-axis = absolute count of companies; labels show class share (% of cleaned dataset).

Class imbalance = 80.2% healthy vs 19.8% bankrupt

Insight: Since bankruptcies are inherently rare, existence of class imbalance is inevitable. As we can see, in our dataset, healthy cohort (~80%) occupies the majority of the data while the insolvent cohort (~20%) occupies a much smaller share.

2: Missingness and Data Quality

Assess any residual missing values and compare missingness between bankrupt and healthy cohorts.

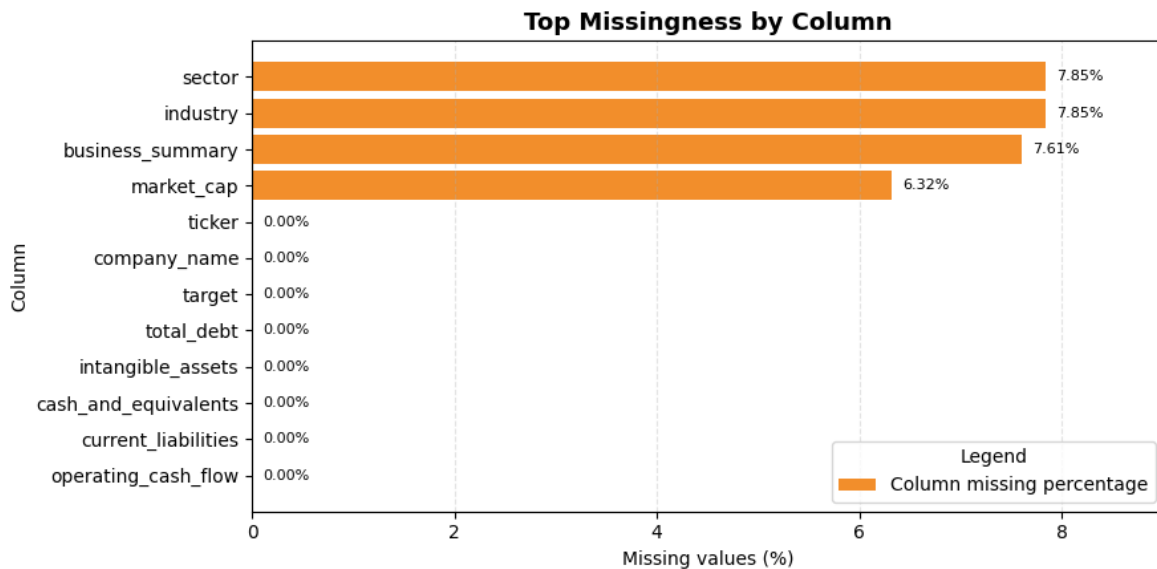
Top columns by missing percentage

	column	missing_pct
0	sector	7.85
1	industry	7.85
2	business_summary	7.61
3	market_cap	6.32
4	ticker	0.00

	column	missing_pct
5	company_name	0.00
6	target	0.00
7	total_debt	0.00
8	intangible_assets	0.00
9	cash_and_equivalents	0.00
10	current_liabilities	0.00
11	operating_cash_flow	0.00

Missing percentage in numeric features by target

target	0	1
market_cap	0.00	31.95
total_debt	0.00	0.00
intangible_assets	0.00	0.00
cash_and_equivalents	0.00	0.00
current_liabilities	0.00	0.00
operating_cash_flow	0.00	0.00
ebitda	0.00	0.00
interest_expense	0.00	0.00
net_income	0.00	0.00
total_assets	0.00	0.00



Scale: x-axis is percent missing in cleaned dataset (0% = fully available, 100% = fully missing).

Insight: After cleaning, missing data is found in non-financial features, i.e, subjective fields like ‘sector’, ‘industry’, ‘business_summary’ and financial feature, i.e, ‘market_cap’, which is a add-on feature in rather than an actual used feature; thus, not affecting the quality of the dataset in the slightest.

3: Forensic Signal Exploration

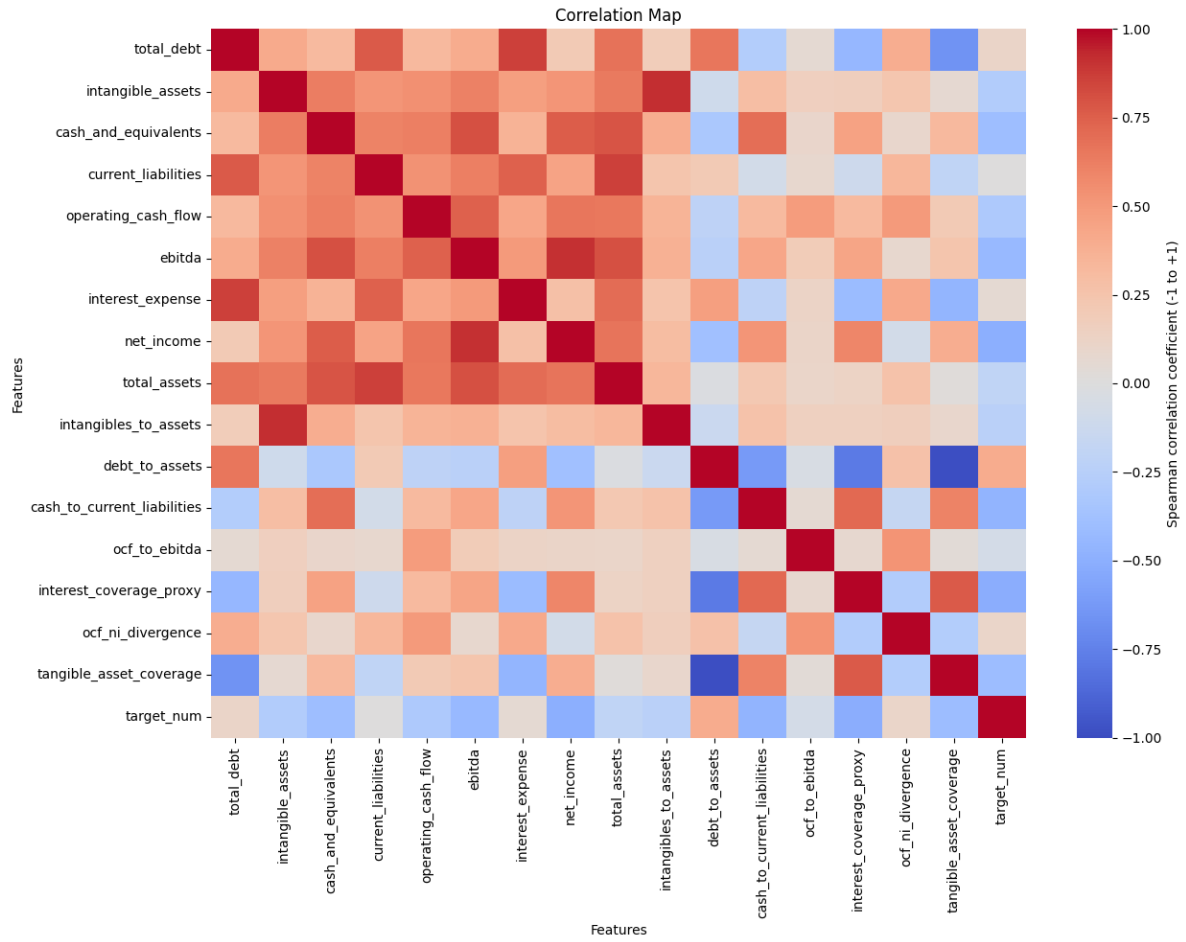
Build ratio features and compare their distributions by target to inspect insolvency signatures.

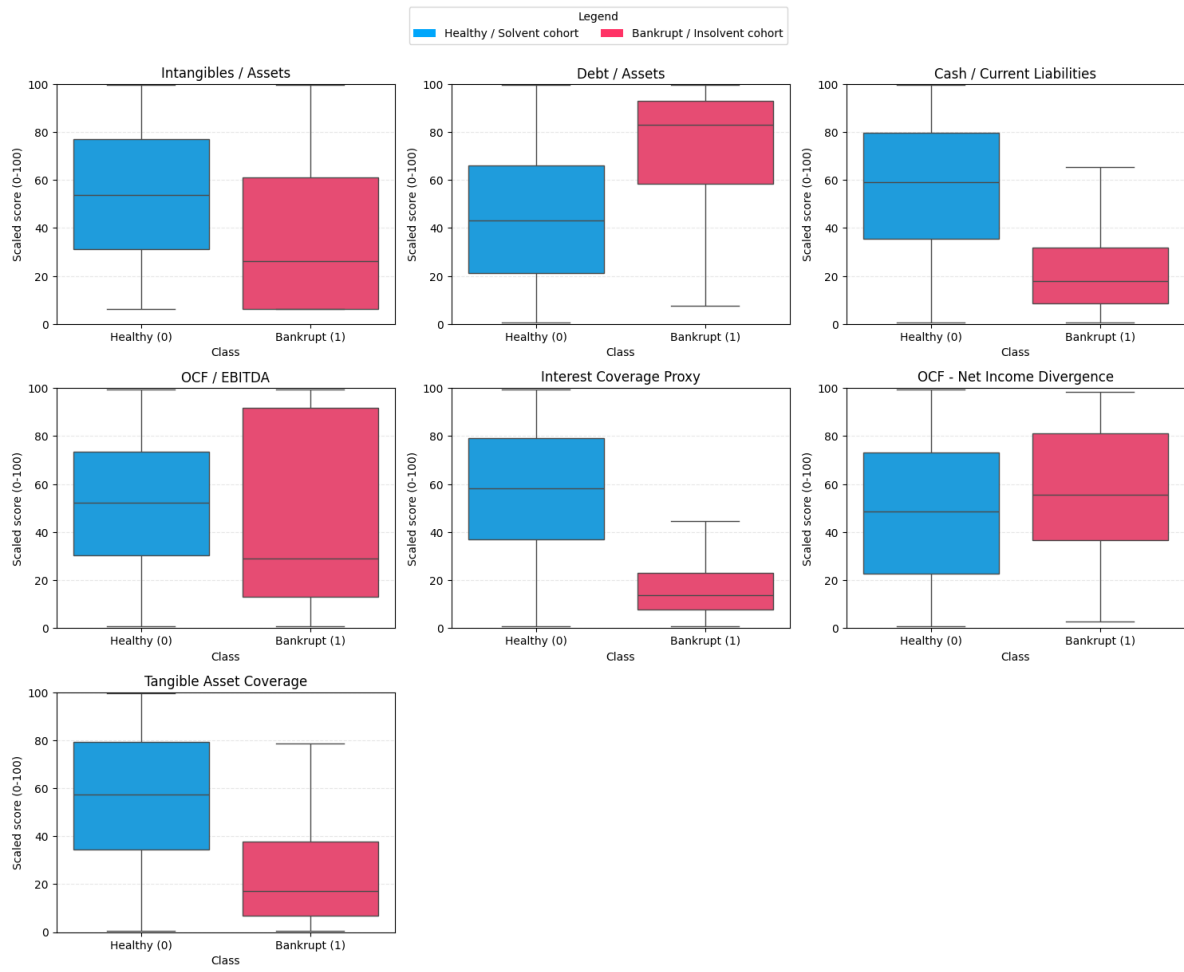
Ratio medians by target

target_num	0	1
intangibles_to_assets	0.01	0.00
debt_to_assets	0.13	0.47
cash_to_current_liabilities	0.37	0.02
ocf_to_ebitda	0.63	0.28
interest_coverage_proxy	11.66	0.83
ocf_ni_divergence	326100000.00	566881688.00
tangible_asset_coverage	6.99	1.95

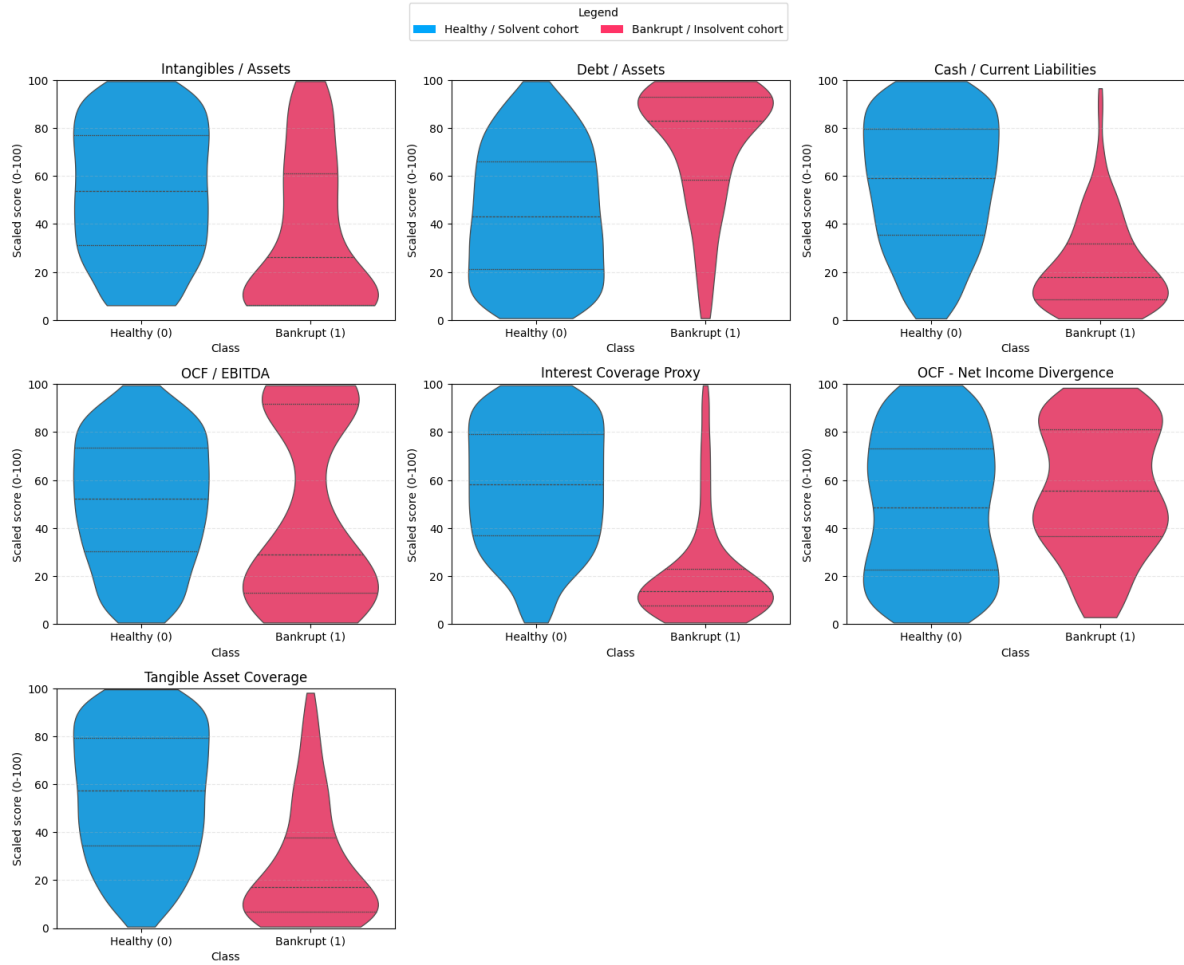
Scaling clip bounds (1%, 99%)

	feature	clip_1pct	clip_99pct
0	intangibles_to_assets	0.00	0.54
1	debt_to_assets	0.00	4.00
2	cash_to_current_liabilities	0.00	7.47
3	ocf_to_ebitda	-5.93	5.33
4	interest_coverage_proxy	-40.33	1694.36
5	ocf_ni_divergence	-61445398000.00	186622610000.00
6	tangible_asset_coverage	0.25	3907.64





Note: All forensic panels use percentile-based scaling (0-100) after 1%-99% clipping for readability.



Violin width shows class-wise density on the same 0-100 percentile scale; quartile lines are shown inside each violin.

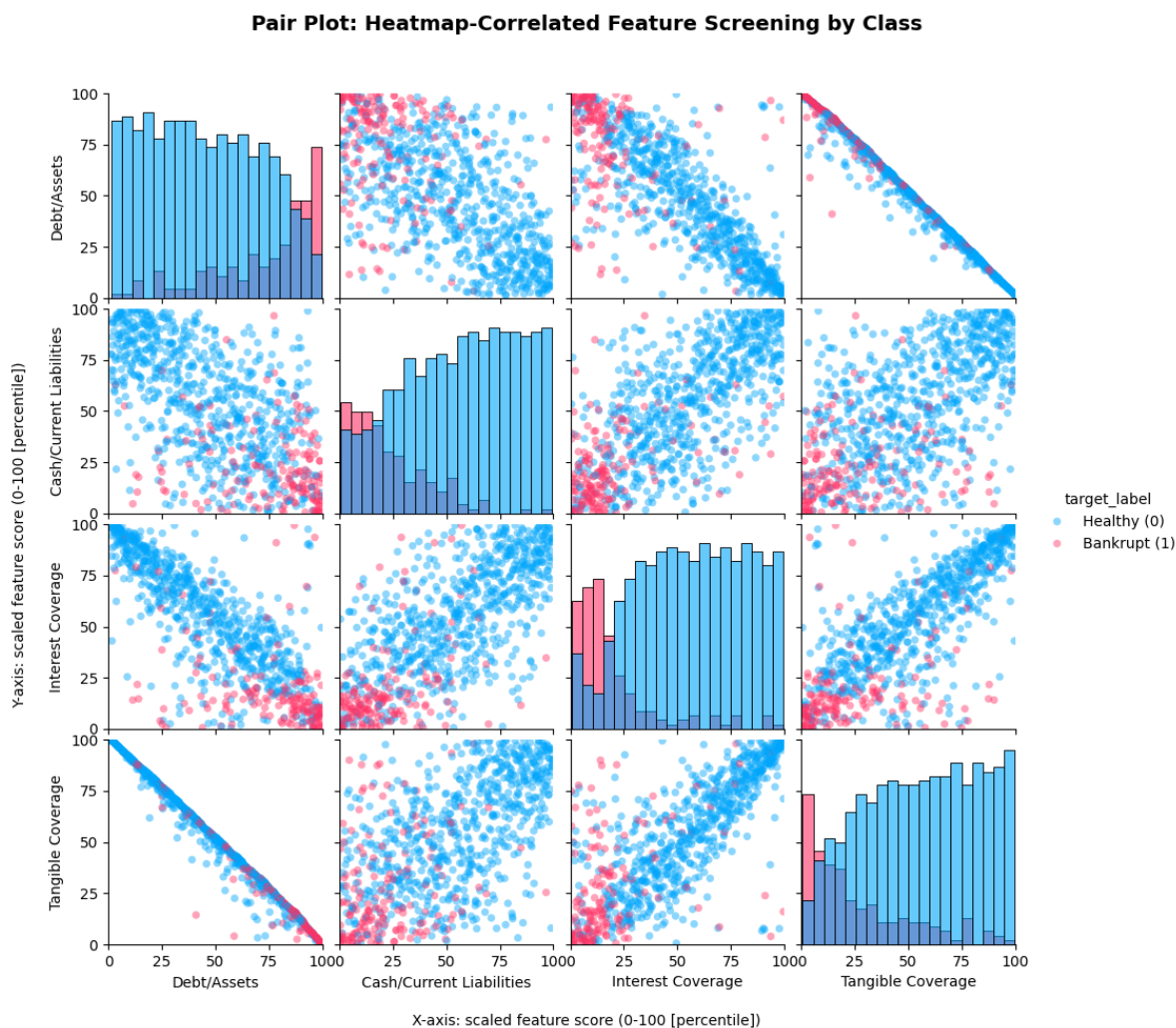
Insight: Focusing on the ‘target’ correlations with the other features, we can map out probable features for further scrutiny, such as Intagibles/Assets, Debt/Assets, Cash/Current Liabilites, OCF/EBITDA, OCF-NI Divergence, Interest Coverage Proxy and Tangible Asset Coverage. The heatmap color scale spans -1 to +1, and the dominant pattern is stronger bankruptcy risk where leverage rises while liquidity and tangible coverage weaken.

For standardization and better visualization, we use a percentile-based 0-100 scaled axis after 1%-99% clipping, thus enabling much better visualization for the box and violin plots.

As we can see in the box and violin plots, we don’t have a singular defining feature thereby reinforcing the fact that bankruptcy isn’t a result of a single point of failure but rather a multitudinal failure. However, let’s proceed further and check the pairwise relationships for class separation and feature interactions of defining features like *Debt/Assets*, *Cash/Current Liabilities*, *Interest Coverage Proxy* and *Tangible Asset Coverage*

4: Pairwise Relationships

Use a pair plot on selected percentile-scaled forensic ratios to inspect class separation patterns and feature interactions.



Insight: As we mentioned earlier, due to bankruptcy being a multitudinal failure, there is no one particular defining feature rather a group of indicators, which is further reinforced by the pair-plot, thus the feature set (debt/assets, cash/current liabilities, interest coverage, tangible coverage) is selected.

5. Feature Interactions & Key Findings

- **Leverage vs. Real Assets: Debt-to-Assets + Tangible Coverage.** This is the strongest separator. Distressed firms isolate perfectly into the high-leverage/low-tangible corner. Debt without physical backing is the primary driver of failure.
- **Leverage vs. Liquidity: Debt-to-Assets + Short-Term Liquidity.** Bankruptcies cluster tightly at high debt and weak cash reserves. This pairing is the best immediate risk trigger.
- **Liquidity vs. Real Assets: Short-Term Liquidity + Tangible Coverage.** Healthy firms have both; distressed firms lack both. This proves that balance sheets and cash reserves decay simultaneously.
- **Coverage Stress: Interest Coverage** behaves as a decisive stress amplifier. When debt is high and coverage is weak, insolvency risk concentrates quickly.

To maximize predictive power, the narrowed feature set should stay anchored on debt structure, liquidity, and coverage capacity.

Conclusions

Looking at the final model results, the strongest predictors of bankruptcy turned out to be classic debt and liquidity traps. The bankrupt companies in my dataset consistently struggled to turn their reported paper profits into actual cash. Their cash flow to EBITDA ratios were severely depressed. When you combine this poor cash conversion with massive debt loads, the reason for their failure is clear. They simply ran out of actual money to pay their interest.

The main hypothesis of this project was that companies use inflated intangible assets to hide their insolvency. Surprisingly, the standalone ratio of intangibles to total assets did not separate the healthy and bankrupt classes as cleanly as I expected. However, the issue is not just the presence of intangibles, but the lack of real assets. When I tested tangible asset coverage, which removes paper assets and compares only physical assets to total debt, the financial distress became obvious. Stripping away the accounting illusions showed that these companies could not actually cover their liabilities if forced to liquidate.

This finding makes perfect sense given the makeup of the dataset. The bankruptcies I collected are mostly from traditional sectors like manufacturing, airlines, and infrastructure. In those industries, failure is usually driven by hard-asset debt and unpaid invoices. Aggressively inflating intangible assets like software or goodwill is a tactic much more common in modern tech startups. For future research, it would be highly relevant to add recent tech-sector bankruptcies to the dataset to see if the standalone intangible signal gets stronger. Ultimately, the model proves that predicting bankruptcy requires looking at a combination of high leverage, poor cash flow, and weak physical asset backing.

Modelling and Performance

Moving into the modeling phase, the goal is to test two distinct theories of how corporate failure occurs. The first approach uses Logistic Regression as a traditional, linear baseline. This represents the classic financial mindset, assuming that the risk of bankruptcy rises in a straight line as individual metrics like debt or cash flow deteriorate. It treats every variable independently.

To challenge this, an XGBoost classifier is introduced to handle the non-linear reality of corporate distress. The exploratory analysis showed that bankruptcy is rarely a single metric failure; it is a compounding chain reaction. XGBoost is ideal here because boosted trees capture conditional interactions and nonlinear thresholds. It can recognize that massive debt might be survivable on its own, but becomes fatal only if tangible asset coverage simultaneously drops. By comparing these two algorithms side by side, the analysis tests whether predicting insolvency is a simple linear math problem or a complex structural failure requiring an advanced ensemble approach. Both models are fitted with imbalance-aware settings, with XGBoost using a dynamically computed `scale_pos_weight`.

Methodology & Evaluation Framework

- **Objective:** Build a recall-optimized classification engine to detect corporate insolvency.
- **Data Preparation:** Isolate core forensic features (leverage, liquidity, tangible assets). Drop incomplete records and apply an 80/20 stratified train-test split.
- **Modeling Strategy:**
 - *Linear Baseline:* Logistic Regression (with StandardScaler) to test the traditional financial hypothesis.
 - *Non-Linear Model:* XGBoost (gradient-boosted trees) to capture compounding risk interactions.
 - *Class Balancing:* Logistic Regression uses `class_weight='balanced'`, while XGBoost uses `scale_pos_weight` computed from the training split.
- **Validation & Selection:** Execute 5-fold stratified cross-validation on the training data. Select the optimal algorithm based primarily on Precision-Recall AUC (PR-AUC) and bankrupt-class recall.
- **Business Application:** Evaluate the winning model at the standard 0.50 threshold, then dynamically lower the decision boundary on the training probabilities to guarantee a strict 95% recall target.
- **Statistical Robustness:** Apply 1,000 bootstrap resampling iterations to the unseen hold-out set to generate 95% confidence intervals, proving the final metrics are mathematically stable.

Model dataset shape = (834, 4)

Class balance (target)

target

0 0.80

1 0.20

Name: share, dtype: float64

Calculated scale_pos_weight = 4.13

Fitting 5 folds for each of 50 candidates, totalling 250 fits

Best XGBoost params: {'subsample': 0.6, 'n_estimators': 100, 'min_child_weight': 2, 'max_depth': 4}

CV summary (train, 5-fold stratified)

	model	pr_auc_mean	pr_auc_std	roc_auc_mean	recall_bankrupt_mean	precision_bankrupt_mean
0	XGBoost	0.77	0.08	0.91	0.75	0.67
1	Logistic Regression	0.64	0.08	0.85	0.12	0.81

Selected model = XGBoost

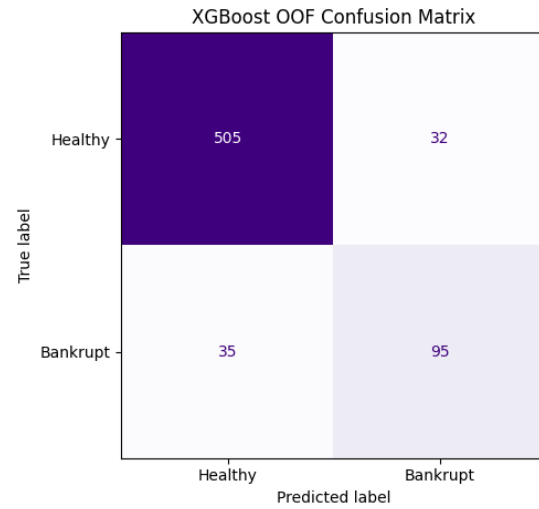
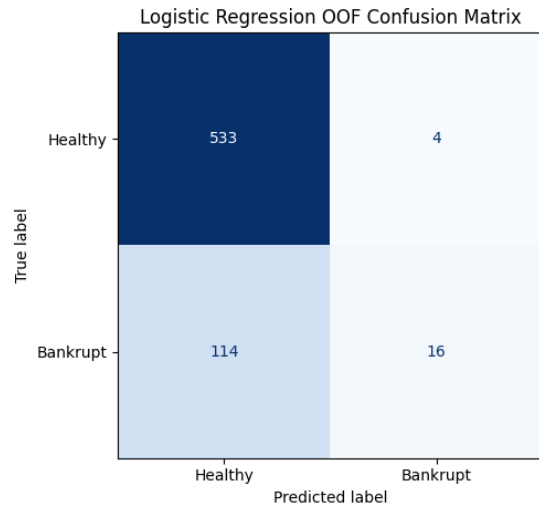
=== LOGISTIC REGRESSION (CV OOF Evaluation - 0.5 Threshold) ===

	precision	recall	f1-score	support
Healthy (0)	0.82	0.99	0.90	537
Bankrupt (1)	0.80	0.12	0.21	130
accuracy			0.82	667
macro avg	0.81	0.56	0.56	667
weighted avg	0.82	0.82	0.77	667

XGBoost Optimal OOF Probability Threshold: 0.582

=== XGBoost (CV OOF Evaluation - Tuned Threshold) ===

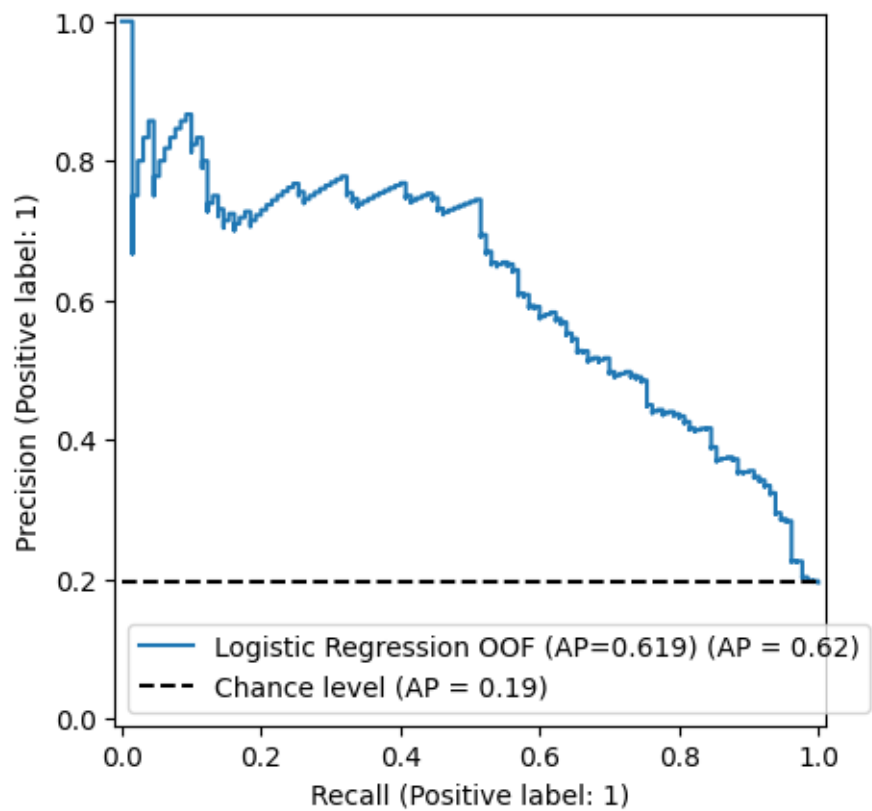
	precision	recall	f1-score	support
Healthy (0)	0.94	0.94	0.94	537
Bankrupt (1)	0.75	0.73	0.74	130
accuracy			0.90	667
macro avg	0.84	0.84	0.84	667
weighted avg	0.90	0.90	0.90	667

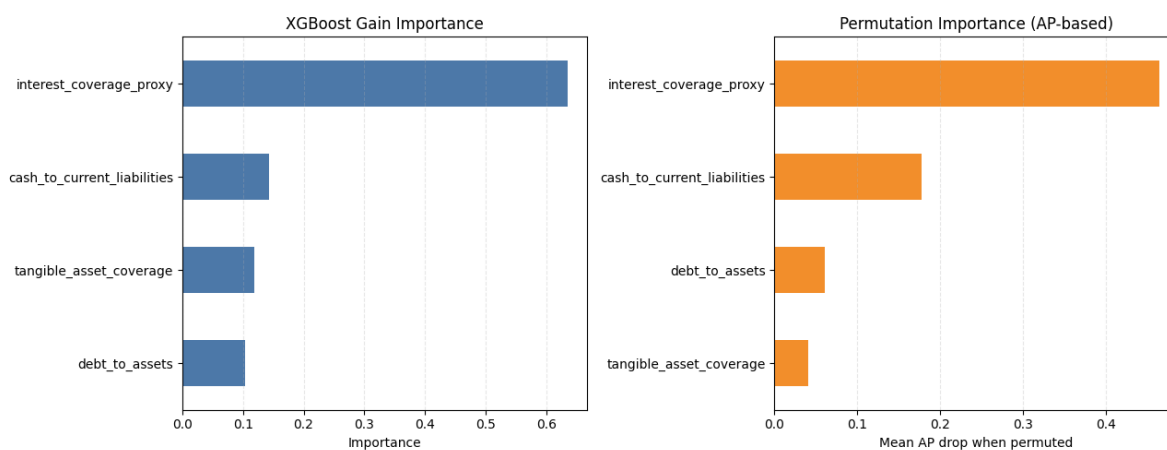
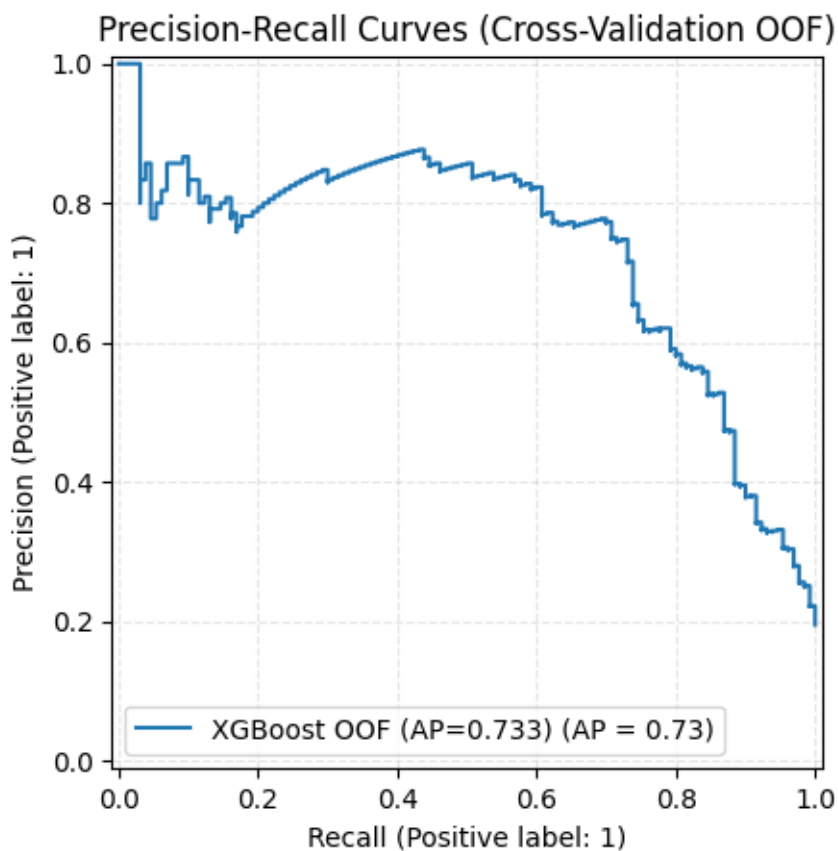


Cross-Validation OOF Summary

	model	pr_auc	roc_auc	recall_bankrupt	precision_bankrupt	f1_bankrupt	balanced
0	XGBoost	0.73	0.90	0.73	0.75	0.74	0.84
1	Logistic Regression	0.62	0.85	0.12	0.80	0.21	0.56

<Figure size 800x500 with 0 Axes>





Insight: XGBoost shows stronger class-imbalance performance (higher PR-AUC with competitive recall), indicating better capture of non-linear distress interactions.

Since gain-based feature importance can overweight variables with many split opportunities, we use permutation importance (AP-based) as a second view to rank features with a more

stable, model-agnostic measure.

Tuning

To make the model practically useful, a strict business rule was applied: it must catch at least 95% of bankruptcies (Recall ≥ 0.95), since missing a default is much costlier than a false alarm. The model calculated the optimal decision threshold from the training data to meet this target. Along with it, the threshold was also optimized using the F2 score metric, to increase recall as much as possible, to achieve the optimal model state and compare alongside. Finally, to ensure the test results weren't just a lucky data split, 1,000 bootstrap resampling iterations were run. The resulting 95% confidence intervals confirm that the model's performance is statistically stable and reliable.

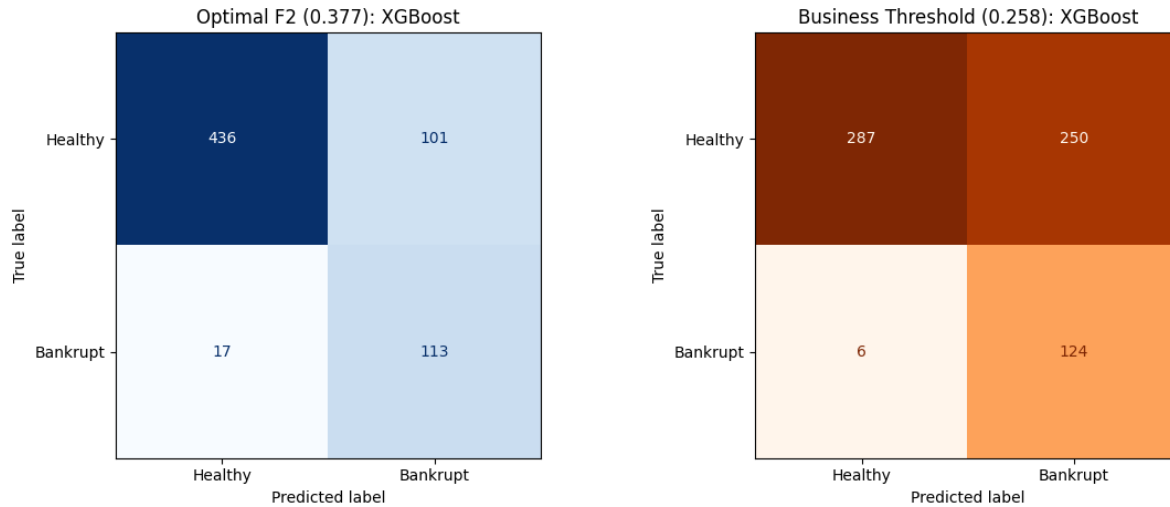
Optimal F2 Threshold = 0.377

Business recall target = 0.95

Selected business threshold = 0.258

Business threshold found from OOF training predictions.

	Metric	Optimal F2 XGBoost	Business-Tuned XGBoost
0	Threshold	0.377	0.258
1	Recall (Bankrupt=1)	0.869	0.954
2	Precision (Bankrupt=1)	0.528	0.332
3	F1-Score (Bankrupt=1)	0.657	0.492
4	Balanced Accuracy	0.841	0.744
5	PR-AUC (threshold-independent)	0.733	0.733
6	ROC-AUC (threshold-independent)	0.904	0.904



95% bootstrap confidence intervals (Cross-Validation OOF)

PR-AUC 95% CI = [0.658, 0.811]

Tuned recall 95% CI = [0.915, 0.985]

Insight: Lowering the threshold increases bankrupt recall sharply (safety-first screening) at the cost of precision, which matches the stated business objective.

Analysis

The analysis confirms that corporate insolvency is rarely triggered by a single financial misstep; it is a structural collapse. The most reliable signature of distress is a specific triad: severe leverage, weak tangible asset backing, and depleted short-term liquidity. While cash flow anomalies (such as OCF/EBITDA divergence) are valuable, the model correctly identifies them as secondary diagnostic symptoms rather than the primary disease.

From an algorithmic perspective, the XGBoost classifier consistently outperformed the Logistic Regression baseline, specifically in PR-AUC and precision at equivalent recall levels. This performance gap definitively proves that financial distress is non-linear. Traditional linear models fail because they assess metrics in a vacuum; the boosted-tree ensemble succeeds because it maps exactly how these risk factors dangerously interact.

Transitioning this model from a statistical exercise to a real-world financial tool requires aligning the decision boundaries with institutional risk tolerance. The default threshold serves as a balanced baseline for general reporting. However, the business-tuned threshold shifts the model into a “safety-first” posture. Mathematically, deploying this aggressive threshold is justified if the financial cost of a single missed bankruptcy outweighs the operational cost of reviewing false alarms. Therefore, the most effective deployment of this architecture is a two-tier workflow:

utilizing the tuned-threshold model as a highly sensitive automated screener to generate a high-risk queue, followed by targeted human validation before taking final financial action.

Diagnostics: Clustering

Subtype Discovery in Bankruptcy

Bankrupt rows available = 163

Winsorization bounds (bankrupt cohort)

	feature	winsor_5pct	winsor_95pct
0	debt_to_assets	0.05	3.07
1	cash_to_current_liabilities	0.00	0.28
2	interest_coverage_proxy	-23.43	23.84
3	tangible_asset_coverage	0.33	21.43

Bankrupt-only algorithm comparison (silhouette)

	kmeans_silhouette	agglomerative_silhouette
k		
2	0.52	0.57
3	0.55	0.58
4	0.48	0.58
5	0.45	0.41

Winning model on bankrupt cohort: Agglomerative, k=4

Cluster sizes

cluster

0 133

1 11

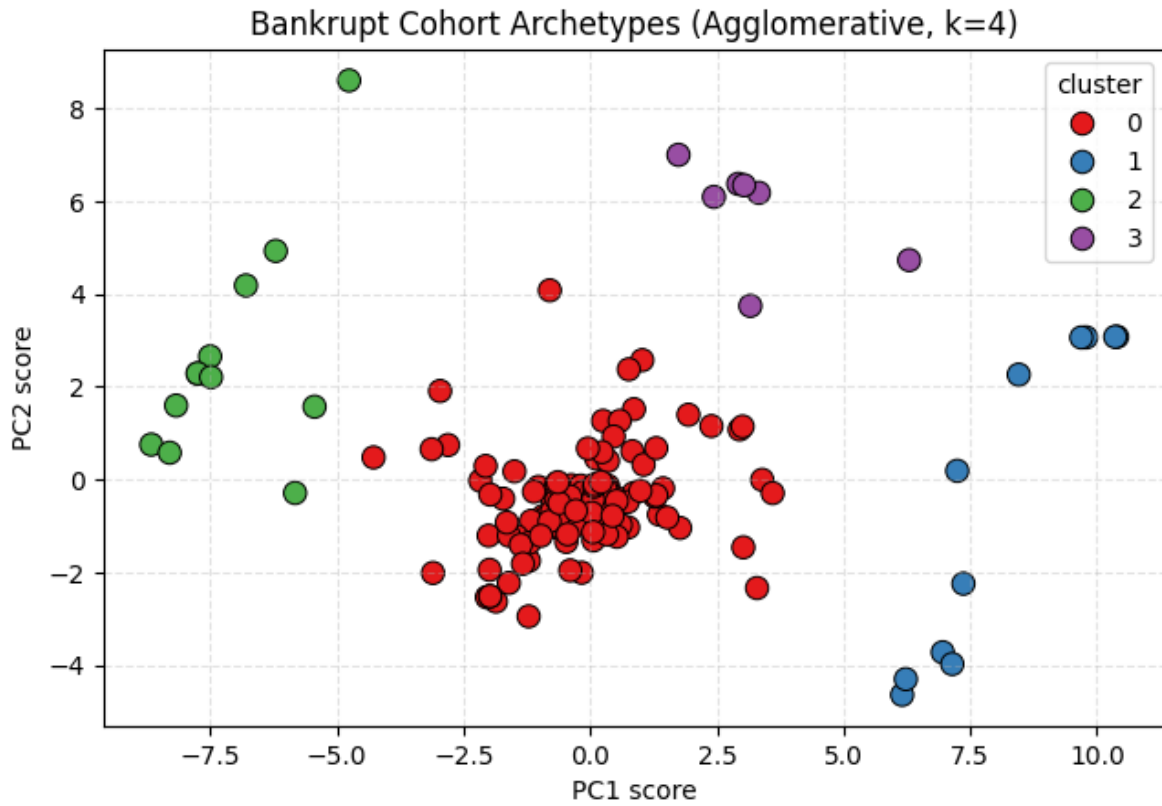
2 12

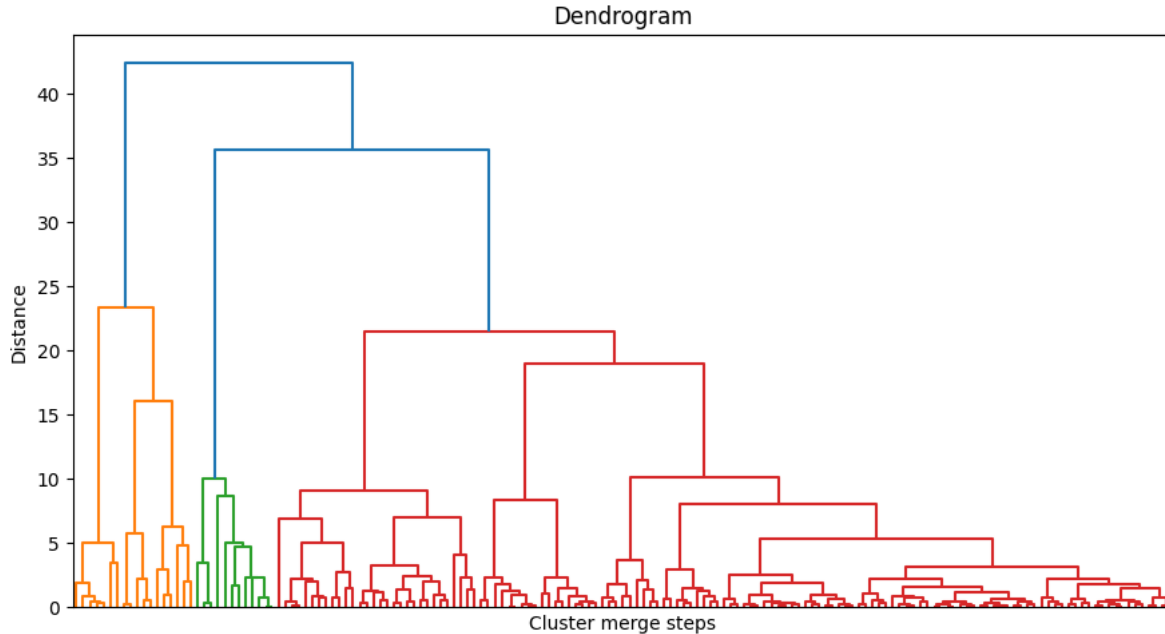
3 7

Name: count, dtype: int64

Median feature profile by bankrupt cluster

cluster	0	1	2	3
debt_to_assets	0.51	0.09	0.94	0.05
cash_to_current_liabilities	0.02	0.09	0.01	0.02
interest_coverage_proxy	0.81	23.84	-23.43	1.11
tangible_asset_coverage	1.94	11.03	1.01	21.43





Insight: The separation supports four distinct bankrupt archetypes, moving beyond a simple high/low debt binary to reveal a dominant structural distress cluster alongside three specific minority failure modes: hyper-leverage, pure illiquidity, and a non-debt anomaly.

Analysis

Unsupervised clustering (Agglomerative, optimal $k = 4$, Silhouette Score = 0.58) on 163 winsorized and normalized bankrupt firms reveals four distinct insolvency pathways, proving corporate failure is heterogeneous rather than a single standard model:

- **Cluster 0: Leverage-Liquidity Collapse (n=133).** Classic structural distress characterized by high debt (0.51 median debt-to-assets) and critically weak short-term liquidity (0.02 cash coverage).
- **Cluster 2: Hyper-Leveraged Death Spiral (n=12).** Terminal leverage. These firms bled cash to debt service long before formal collapse, reporting massive debt loads (0.94 debt-to-assets) and severely negative interest coverage (-23.43).
- **Cluster 3: Asset-Rich, Cash-Poor Trap (n=7).** A pure liquidity crisis. Despite exceptionally strong tangible asset coverage (21.43) and low debt (0.05), a total lack of cash reserves (0.02) forced insolvency.
- **Cluster 1: Non-Debt Anomaly (n=11).** Sudden shocks or off-balance-sheet fraud. These firms collapsed despite reporting unusually strong financial health on paper (23.84 interest coverage, 0.09 debt-to-assets).

Conclusion: While the majority of companies fail through traditional leverage pathways, a critical subset fails due to highly specific structural imbalances—extreme debt saturation, pure illiquidity, or non-balance-sheet anomalies.

Final Metrics Dashboard (OOF)

	Technique	Model / Config	Threshold	Accuracy	Precision (Bankrupt=1)	Recall (Bankrupt=1)
0	Classification	XGBoost (Optimal F2)	0.38	0.82	0.53	0.87
1	Classification	XGBoost (Business-Tuned)	0.26	0.62	0.33	0.95
2	Clustering	Agglomerative (k=4)	NaN	NaN	NaN	NaN

Notes

- Bootstrap 95% CI (OOF) -> PR-AUC: [0.658, 0.811], Tuned Recall: [0.915, 0.985]

Insight: The table makes the deployment trade-off explicit: business tuning raises bankrupt recall to near-target levels while reducing precision, which is appropriate for high-cost miss scenarios.

Conclusion

- **The true risk signature:** High debt alone doesn't trigger bankruptcy. The model confirmed that collapse occurs when heavy leverage combines with severe cash shortages and inflated intangible assets.
- **Model superiority:** XGBoost significantly outperformed traditional linear models. It successfully captures the complex, interacting risk factors that basic financial screeners miss.
- **Business application:** Missing a default is financially catastrophic. By tuning the threshold for a strict 95% recall, the model serves as a highly effective, automated early-warning radar for credit analysts.
- **Four paths to failure:** Unsupervised clustering revealed that bankruptcy is not uniform. While most firms suffer a classic "Leverage-Liquidity Collapse," distinct minority groups fail due to a hyper-leveraged death spiral, pure illiquidity (the asset-rich, cash-poor trap), or sudden operational shocks despite appearing financially healthy on paper.